

ALIGHT INTERNATIONAL MARKETPLACE

PRODUCTION GRADE SPECIFICATION

Stage 13: Payments, Multi-Currency, Escrow & Webhook Processing

1. Overview & Architecture

Stage 13 implements a secure, resilient, multi-gateway payment infrastructure supporting multi-currency calculations, asynchronous webhook ingestion with cryptographic signature validation and idempotency deduplication, automated escrow holding for multi-vendor sub-orders, refund reversals, and offline bank transfer manual clearance workflows.

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```
flowchart TD
    Buyer[Buyer Checkout] -->|Initiate Payment| PS[PaymentService]
    PS --> GW[PaymentGatewayFactory\nRazorpay / Stripe / Bank Transfer / Mock]
    GW --> PGW[(External Payment Gateway)]

    subgraph Async Webhook Flow
        PGW -->|Webhook Callback| PWC[PaymentWebhookController]
        PWC --> PWS[PaymentWebhookService]
        PWS -->|Idempotency Check| DB_WH[(payment_webhooks)]
        PWS -->|Capture Transaction| PT[(payment_transactions)]
        PWS -->|Hold in Escrow| SS[SettlementService]
        SS --> VW[(vendor_wallets)]
        PWS -->|Publish Event| PSE[PaymentSucceededEvent]
    end

    subgraph Multi-Currency Engine
        CS[CurrencyService] -->|Base Rate Cross-Calculation| CURR[(currencies & exchange_rate_history)]
        CS -->|Conv[Live FX Rate Conversion]
    end
```

2. Key Components & Implementation Details

A. Database Schema Migration (`V39__payments_multi_currency_and_webhooks.sql`)

- Enhanced `payment_webhooks` table:
 - `signature VARCHAR(255)` : Cryptographic signature snapshot. - `ip_address VARCHAR(45)` : Source client IP address. - `error_message TEXT` : Processing error details. - `retry_count INT` : Re-delivery attempt counter. - `updated_at TIMESTAMP WITH TIME ZONE` : Audit timestamp.
- Performance Indexes:
 - `idx_payment_webhooks_gateway_created` on `payment_webhooks(gateway_type, created_at DESC)` -
 - `idx_payment_tx_gateway_order` on `payment_transactions(gateway_order_id)` -
 - `idx_payment_tx_composite_status` on `payment_transactions(order_id, transaction_status)` -

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idx_exchange_rate_history_pair on exchange_rate_history(from_currency_id, to_currency_id,
created_at DESC)
```

B. Multi-Gateway Payment Adapters & Initiation

- **Gateway Adapters (`gateway/`):**
 - `RazorpayGatewayAdapter` : Order creation, HMAC-SHA256 signature verification, and instant refund processing.
 - `StripeGatewayAdapter` : PaymentIntent creation and charge capture.
 - `BankTransferGatewayAdapter` : B2B NEFT/RTGS virtual account generation and manual clearance submission.
 - `MockGatewayAdapter` : Zero-dependency testing and staging mock simulations.
- **Payment Lifecycle (`PaymentServiceImpl.java`):**
 - Tracks state transitions: `INITIATED` → `AUTHORIZED` → `CAPTURED` → `REFUNDED` / `FAILED` . - Publishes `PaymentSucceededEvent` upon capture.

C. Asynchronous Webhook Receiver & Idempotency Engine

- **Webhook Service (`PaymentWebhookServiceImpl.java`):**
 - Validates webhook idempotency using `existsByEventId(eventId)` . - Parses webhook payloads into gateway order and payment identifiers.
 - Captures corresponding `PaymentTransaction` and marks `Order` as `PAID` and `CONFIRMED` . - Automatically holds sub-order funds in vendor wallets under `pending_balance` (Escrow).
 - Webhook controller endpoints:
 - `POST /api/v1/payments/webhooks/razorpay` (`X-Razorpay-Signature`)
 - `POST /api/v1/payments/webhooks/stripe` (`Stripe-Signature`)
 - `POST /api/v1/payments/webhooks/generic/{gatewayType}`

D. Multi-Currency Live FX Engine (`CurrencyServiceImpl.java`)

- Dynamic cross-rate calculation via platform base currency (INR default).
- Support for variable decimal places (e.g. 0 for JPY, 2 for USD/EUR/INR).
- Exchange rate update auditing with historical tracking in `exchange_rate_history` .

E. Refund & Escrow Reversal Orchestration

- Automated full and partial refund processing via gateway adapters.
- Reverses pending escrow allocations in vendor wallets upon refund approval.

3. Verification & Test Suite

- Comprehensive end-to-end integration test suite implemented in `PaymentAndMultiCurrencyIntegrationTest.java` :
 1. `testMultiCurrencyConversionsAndExchangeRates` : Multi-currency conversions, base rate cross-calculations, formatting, and exchange rate auditing.
 2. `testPaymentInitiationAndEscrowHold` : Multi-gateway payment initiation, signature verification, and automatic escrow placement in vendor wallet.
 3. `testWebhookProcessingAndIdempotency` : Webhook callback processing, transaction capture, escrow allocation, and duplicate deduplication.
 4. `testRefundProcessingAndEscrowReversal` : End-to-end refund processing and escrow reversal.
 5. `testBankTransferApprovalWorkflow` : B2B offline bank transfer submission and administrative clearance approval.

- Unit test suite: `PaymentServiceTest.java`.
- Build compilation: `BUILD SUCCESS` across 57 test classes.